IE 313
SUPPLY CHAIN MANAGEMENT
Project Report

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1. Introduction

This study analyzes the aggregate production planning problem of Q&H, a company that is observed to have fluctuating demand over a period of one year. A mathematical model is formulated to solve the problem and obtain an optimal solution while considering various factors such as workforce costs, material costs, and holding costs. The main goal is to minimize costs and maximize profits for the company. The analysis is further extended to cover the impact of promotional pricing strategies on the demand and profits for the company. The study considers different scenarios for promotional pricing and the impact of a rival company named Unilock. The results obtained from the model help in understanding the impact of promotional pricing strategies.

2. Model Implementation

The aggregate production planning problem was modeled in Python using the Pyomo optimization library and solved with the GLPK solver. Two model structures were developed: one for cost minimization and another for profit maximization. In both models, monthly production, subcontracting, and inventory levels were defined as decision variables, while demand and price values were introduced as parameters. The objective function either minimizes total operational costs or maximizes profit depending on the scenario considered. The model includes constraints for production capacity, inventory balance, safety stock requirements, and the initial and final inventory levels.

Different promotional and competitive scenarios were analyzed by modifying the demand and price parameters and resolving the model for each case. This structure allows the same optimization model to be reused across different scenarios while evaluating the impact of promotion timing and competitor actions on the optimal production plan and profitability.

3. Aggregate Production Planning and Cost Optimization

The aggregate production planning model was initially solved with the aim of minimizing the total cost over the 12-month period. The optimal solution resulted in a total cost of $9,112,500. From the production plan, it can be noted that internal production remains constant at 160 tons per month, which reflects the capacity constraint of the production line due to the fixed workforce and hours of operation. Since this capacity is not enough to meet the variable monthly demand with the required safety stock level, subcontracting occurs every month. The subcontracting varies depending on the demand level, with much higher levels occurring in months with high demand levels, such as April and December. Inventory levels are kept at the minimum safety stock level of 100 tons for most months to reduce inventory costs, with inventory levels increased to 250 tons in the last month to meet the requirement of having the same inventory levels at the end of the planning horizon as those at the beginning.

4. Profit Maximization Analysis

In the second part of the analysis, the goal of the model is changed to maximize profits, where profits are calculated as total revenue minus total cost. As the price per unit of detergent sold is fixed at $2,800 per ton, the model aims to find the solution that maximizes the total profits, considering the constraints on meeting the total demand and the operation constraints. The optimal solution indicates that the total profits are $3,470,700.

The optimal solution for the production level remains the same as in the previous case, where internal production is at the maximum possible level of 160 tons per month, considering the capacity constraint, and subcontracting is used to meet the total demand. The level of inventory is kept at the safety stock level of 100 tons for most of the planning period and then increased to 250 tons in the last month to meet the required inventory level. The solution indicates that in the case where profits are maximized, and the price per unit of detergent sold is fixed, the internal capacity is fully utilized, and strategic subcontracting is employed to meet the total demand, considering the inventory levels.

5. Impact of Promotional Pricing and Demand Timing

Promotions influence the demand and profitability of the product depending on the timing of the promotion. The promotional pricing strategy lowers the selling price from $2,800 to $2,520 per ton for the selected month. Consequently, demand during the promotion month increases by 40%. Moreover, an additional 20% of demand from each of the next two months shifts to the month of promotion due to forward buying.

5.1. April Promotion Dynamics:

If the promotional pricing strategy is adopted in April, the optimal profit increases to $3,571,700. The production schedule still maintains the internal production at a constant level of 160 tons/month, considering the internal production capacity. Due to the increase in demand in April, the level of subcontracting increased to 830 tons in that month. Consequently, as part of the demand in May and June shifts to April due to forward buying, the level of subcontracting in May and June decreases compared to the base case. Similarly, the inventory level is kept at the safety stock level of 100 tons for the entire year, except for the last month in which the required ending inventory level of 250 tons is achieved.

5.2. August Promotion Dynamics:

When the promotion is carried out in August, the optimum level of profit is $3,479,700. As in the previous scenarios, the internal production remains the same, 160 tons per month, and subcontracting is used to meet the increased demand. The promotion will generate a surge in demand in August, and hence, the level of subcontracting in August will be 295 tons. As a result of forward buying, the demand in September and October will be lower, and hence, the level of subcontracting in these two months will be lower than in the case of the original demand scenario.

From the scenarios, it is concluded that the promotion in April generates a higher level of profit than the promotion in August. Although the price in August is lower than in April, the promotion in April enables the company to capture the increased demand early in the planning period, thus leading to an increased level of profit.

6. Competitive Market Analysis

The detergent market is shared by Q&H and its competitor, Unilock. Therefore, the profitability of promotion strategies depends not only on the company's decision to implement a promotion strategy but also on the actions of its competitor. Various promotion scenarios are analyzed to determine the effect on the profitability of both companies.

6.1. Symmetrical and Independent Actions:

If Unilock decides to promote its product in the month of April, while Q&H decides to maintain its prices constant throughout the year, the new profit for Q&H will be $2,997,950. This decrease in profits for Q&H occurs because Unilock's promotion attracts a large part of the market's demand for the product during the promotion month.

If Q&H decides to promote its product in the month of April, while Unilock decides not to promote its product, Q&H's profits will be $3,571,700, and Unilock's profits will be reduced to $2,997,950.

However, if Q&H promotes in August while Unilock does not promote, then the profit is $3,479,700 for Q&H, while the profit is $3,295,200 for Unilock. Thus, the promotion gives Q&H a demand advantage over Unilock, though the advantage is slightly lower compared to the case where the two firms promoted their products in April.

If both firms promote their products in April, then the profit is $3,252,300 for each of the firms. In this case, neither of the two firms has a demand advantage over the other. Both firms only experience the forward buying effect in the following months. When both firms promote their products in August, the profits obtained by the two firms are equal, $3,357,650.

Overall, the results reveal that the highest profit is obtained by Q&H when the firm promotes, while Unilock does not. However, the decision not to promote while the competitor promotes leads to a significant loss in profit due to the low market demand.

6.2. Asymmetric Promotion Timing:

When promotions are implemented in different months by the two firms, the results change. If Q&H promotes in April and Unilock promotes in August, then the profit for Q&H would be $3,396,200, and the profit for Unilock would be $3,006,950. In the above scenario, the competitive advantage for Q&H would be the demand and forward buying effects, as they would promote their products earlier than the competitor's promotion period.

On the other hand, if Q&H promotes their products in August and Unilock promotes their products in April, then the profit for Q&H would be $3,006,950, and the profit for Unilock would be $3,396,200. In the above scenario, the competitive advantage for Unilock would be the demand and forward buying effects, as they would promote their products earlier than the competitor's promotion period.

The results show that the firm that promoted earlier in the year has a competitive advantage as it can capture a greater demand through the consumption effect and the forward buying effect.

7. Strategic Decision Modeling Under Uncertainty

7.1. Collaborative Strategy and Antitrust Regulations:

If Q&H were to coordinate its decisions with Unilock to maximize their collective benefits, the optimal strategy would be to stagger their promotional periods so they do not overlap. According to the profit models, when one firm promotes in April and the other in August, the firm promoting in April achieves a profit of $3,396,200, while the firm promoting in August earns $3,006,950. By coordinating, they can avoid the mutually destructive both promote in the same month scenario, which drops profits to $3,252,300 (for April) or $3,357,650 (for August).

However, this type of coordination is strictly illegal under antitrust laws and competition regulations. Directly or indirectly coordinating promotion timings and pricing is legally defined as "collusion" or "price-fixing". Such agreements hinder free-market competition, artificially manipulate market demand, and ultimately harm consumer welfare. Therefore, while this strategy appears mathematically superior for the firms' total profits, it is not a viable business decision due to the risk of severe administrative fines and legal sanctions.

7.2. The Legal Path: Maximin Strategy:

At this stage, the analysis examines the strategic options available to Q&H under a scenario of uncertainty in which it cannot anticipate the moves of its competitor, Unilock. The company faces two primary options, both of which are shaped by ethical and legal constraints. If the firm wishes to manage its risk while remaining within legal and ethical boundaries, it should apply the maximin strategy from game theory. This approach ensures that the firm safeguards its own profit even in the worst-case scenario against the most aggressive move the competitor could make.

Analysis shows that if Q&H does not run a promotion, profit could drop to $2,997,950. However, if Q&H runs a promotion in April, the lowest possible profit is fixed at $3,252,300. Therefore, the most rational and legal decision that can be implemented without any secret agreement with the competitor is to run a promotion in April.

7.3. Illegal Path and Comparative Assessment:

If the company disregards legal risks and focuses solely on profit maximization, it may enter into a secret coordination with a rival firm. In this scenario, the two firms could coordinate their promotional periods in a way that does not harm each other. Although such a secret agreement could boost the firms' profits to higher levels, it is legally defined as price-fixing. Since it directly hinders free-market competition and consumer welfare, it is subject to severe administrative fines and sanctions under antitrust laws. Consequently, legal risks and ethical violations render this option unsustainable as a business strategy.

In conclusion, both options present different risk and return profiles for Q&H. While the illegal coordination strategy may seem like the most attractive option due to the market share gained through collaboration with the competitor, the severe legal penalties and risk of reputational damage it entails jeopardize these gains. On the other hand, the legal Maximin strategy, while not promising the firm the highest theoretical profit, establishes the safest profit floor ($3,252,300) independent of the competitor's moves. From the perspective of commercial sustainability and ethical business principles, the marginal profit increase offered by the illegal option falls far short of the costs that a potential antitrust investigation would entail. Therefore, the soundest decision for Q&H is the legal April promotional plan, which minimizes external risks.

8. Conlusion

As part of this study, a comprehensive Integrated Production Planning analysis was conducted to manage the demand fluctuations encountered by Q&H over a one-year planning period. The problem was modeled using linear programming techniques, and all analyses were performed using the Python programming language via the Pyomo library.

The operational findings obtained during the analysis process clearly revealed the firm's current capacity constraints and outsourcing needs. It was determined that the firm's monthly internal production capacity of 160 tons was insufficient to meet fluctuating demand, necessitating the strategic use of subcontracting each month. In the base scenario where total costs were minimized, an optimal cost of $9,112,500 was achieved, while inventory levels were observed to be maintained at the safety stock level of 100 tons in most months to reduce costs.

In the section examining promotional strategies, the critical impact of timing on profitability was validated through our Python-based model. Tests revealed that the price discount implemented in April generated a higher profit ($3,571,700) compared to August, as it attracted demand in the early stages of the planning period and provided a market advantage. This demonstrates that promotions not only increase sales volume but also fundamentally alter production and outsourcing planning through the forward buying effect.

Finally, it was found that in scenarios involving the rival firm Unilock, profitability depends not only on the firm's own decisions but also closely on the rival's moves. While the Maximin strategy appears to be the safest legal option under conditions of uncertainty, it has been understood that establishing a secret coordination with the rival could theoretically boost profits even further. However, the significant legal risks associated with illegal coordination options render this option ineligible as a strategic alternative. Consequently, for Q&H, adopting the Maximin approach alongside a risk-averse April promotional plan that remains within legal boundaries is considered the soundest strategic choice.